

Abstract No.

Human Glioma-Infiltrated Brain Slices Reveal Payload-Specific Remodeling by Cytokine-Armored CAR-PBMC

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Background:

CAR immune-cell therapies for glioblastoma remain limited by insufficient activity within the immunosuppressive tumor microenvironment and by the lack of human tissue models that preserve tumor–brain interactions. Human organotypic brain slices provide an opportunity to evaluate engineered immune-cell therapies in a structurally preserved human microenvironment containing tumor cells and non-malignant brain-resident populations.

Methods:

We established a human organotypic glioma-infiltrated brain-slice model by implanting fluorescent patient-derived glioblastoma cells into human cortical slices. Tumor-infiltrated slices were treated with allogeneic PBMC or rapidly generated CAR-PBMC co-electroporated with activating cytokines. Live imaging was used to monitor tumor outgrowth and immune-cell infiltration.

Immunofluorescence assessed tumor-cell apoptosis, invasion-associated markers, and preservation of non-malignant brain-cell populations, including neurons, astrocytes, and microglia. Single-cell transcriptomics were used to define tumor-cell states, immune-cell composition, and payload-dependent remodeling of the tumor–immune microenvironment.

Results:

Cytokine-armored CAR-PBMC infiltrated tumor-bearing human brain slices and reduced glioma outgrowth compared with untreated and mock-PBMC controls ($P < 0.01$). Treatment increased tumor-cell apoptosis, supported by elevated caspase-3 staining (Kruskal–Wallis $P = 0.015$) and apoptosis-associated transcriptional programs. CAR-PBMC also altered tumor architecture and invasion-associated features, including significant reduction of MMP9 (Kruskal–Wallis $P = 0.0070$) and a trend toward reduced Vimentin signal. Non-malignant brain-cell populations were broadly preserved, including astrocytes, microglia, and total nuclei. Different cytokine activation CAR-PBMC had different effects on the neuron populations, from not reduction to modest neuronal reduction. Immune profiling revealed distinct payload-dependent remodeling. Cytokine-armored CAR-PBMC promoted cytotoxic activation but were associated with higher activation, proliferation, regulatory, and lower exhaustion-associated signatures in T-cell programs.



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Conclusion:

Human glioma-infiltrated organotypic brain slices provide a translational platform for evaluating engineered immune-cell therapies within a preserved human tumor–brain microenvironment. Using this model, we show that cytokine-armored CAR-PBMC can restrict glioma growth, induce tumor-cell apoptosis, and reshape the tumor–immune landscape in a payload-specific manner. These findings nominate CAR-PBMC for glioblastoma immunotherapy.



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